

106. If it is in a windless situation, I think both can be used. But if there is a steady wind, I think we should find the average of  $\frac{d}{t_1}$  and  $\frac{d}{t_2}$ . Because:

$$\begin{aligned} \text{1)} \quad V_1 = \frac{d}{t_1} + V_w \quad \frac{d}{t_1} &= V_c + V_w \quad \frac{d}{t_2} = V_c - V_w \\ \Rightarrow \left( \frac{d}{t_1} + \frac{d}{t_2} \right) \times \frac{1}{2} &= \frac{V_c + V_w + V_c - V_w}{2} = V_c \\ \text{2)} \quad \frac{d}{\frac{t_1 + t_2}{2}} &= \frac{2d}{t_1 + t_2} = \frac{d}{V_c t_1 t_2} \end{aligned}$$

$$\begin{aligned} \text{29. a)} \quad 2ax = V_m^2 &\Rightarrow x \approx 10.59 \text{ m} \\ V_m &= 30 \text{ f/m/min} \approx \frac{61}{12} \text{ m/s} \end{aligned}$$

$$\text{b)} \quad 2x + d = L \Rightarrow d \approx 168.82 \text{ m}$$

$$\{ at_1 = V_m \mid t_1 \approx 4.17 \text{ s}$$

$$d = V_m t_2 \Rightarrow t_2 \approx 33.21 \text{ s}$$

$$t = t_1 + t_2 + t_1 \approx 41.55 \text{ s}$$

$$34. \quad V_1 = 20 \text{ km/h} \approx 5.56 \text{ m/s}$$

$$V_2 = 40 \text{ km/h} \approx 11.11 \text{ m/s}$$

$$\{ V_1 t_1 = x_1$$

$$V_2 t_2 = x_2$$

$$x_1 = x_2 + (V_0 t_1 + \frac{1}{2} a t_1^2) \Rightarrow \begin{cases} t_1 \approx 8.06 \text{ s} \\ t_2 \approx 7.01 \text{ s} \\ a \approx 3.29 \text{ m/s}^2 \end{cases}$$

$$x_2 = x_2 + (V_0 t_2 + \frac{1}{2} a t_2^2) \Rightarrow \begin{cases} V_0 \approx 8.74 \text{ m/s} \end{cases}$$

$$47. \quad 2gh = V_x^2 - V_0^2$$

$$\text{a)} \quad 2gh = V_x^2 - V_0^2 \Rightarrow V_x \approx 46.0 \text{ m/s}$$

$$\text{b)} \quad h = \frac{1}{2} g t^2 - V_0 t \Rightarrow t \approx 6.12 \text{ s}$$